

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Project Name	Employee Raise Issue – Record Producer & Service Portal Widget
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

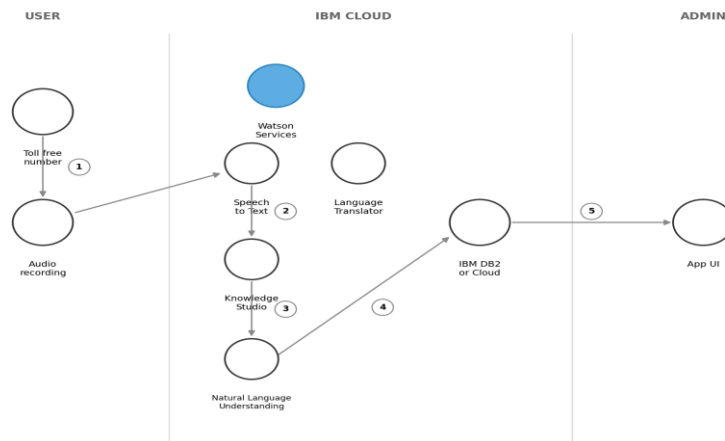


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee Center Service Portal page and widgets used to raise and view issues	HTML, CSS, AngularJS (Service Portal widget framework), Bootstrap
2.	Application Logic-1	Record Producer form logic, variable-to-field mapping	ServiceNow Glide (server-side JavaScript)
3.	Application Logic-2	Category → Subcategory dependency, mandatory field validation	Client Scripts & UI Policies (client-side JavaScript)
4.	Application Logic-3	Auto-assignment and priority calculation on issue creation	Business Rules (server-side JavaScript)
5.	Database	Storage for the custom "Employee raise issue" table	ServiceNow platform database (MySQL-based)
6.	Cloud Database	Hosted instance database service	Now Platform (ServiceNow Cloud)
7.	File Storage	Storage for attachments added to an issue	ServiceNow Attachment API
8.	External API-1	Sending state-change notifications to employees	ServiceNow Email/Notification API
9.	External API-2	Not used in this phase	—
10.	Machine Learning Model	Planned Knowledge Base suggestions / Virtual Agent (future phase)	ServiceNow Virtual Agent (planned)
11.	Infrastructure (Server / Cloud)	Application deployment	Now Platform SaaS instance (ServiceNow Cloud)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source	Bootstrap used for widget styling on the Service Portal	Bootstrap
2.	Security Implementations	Role-based ACLs on the table and portal, HTTPS for all traffic	ServiceNow ACLs, HTTPS/TLS
3.	Scalable Architecture	Built on ServiceNow's scoped application model, isolated from other apps on the instance	ServiceNow scoped application
4.	Availability	Now Platform SaaS provides high-availability infrastructure with vendor-managed uptime	Now Platform SLA
5.	Performance	Client-side UI Policies and Client Scripts reduce server round-trips for validation	Client Scripts, UI Policies

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>